EXERCISE 11

Solution 1:

(i) No, it doesn't lie on the same base and between the same parallels.

(ii) No, it doesn't lie on the same base and between the same parallels.

(iii) Yes, it lies on the same base and between the same parallels. The same base is AB and the parallels are AB and DE.

(iv) No, it doesn't lie on the same base and between the same parallels.

(v) Yes, it lies on the same base and between the same parallels. The same base is BC and the parallels are BC and AD.

(vi) Yes, it lies on the same base and between the same parallels. The same base is CD and the parallels are CD and BP.

Solution 2: Construct AM ⊥ DC and CL ⊥ AB. Extend the lines DC and AB and join AC.

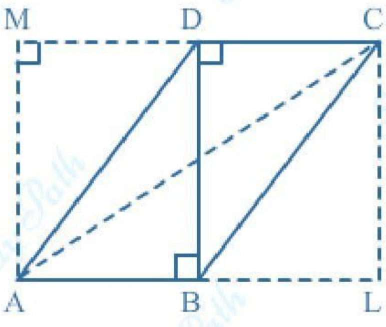
We know that

Area of the quadrilateral ABCD = area of triangle ABD + area of triangle DCB

⇒ar (quadrilateral ABCD) = 2 (Area of △ABD) [Since, ar(△ABD) = ar(△DCB)]

$\therefore ar\ (\triangle \text{ ABD}) = \frac{1}{2}(\text{area of quadrilateral ABCD}) \ \dots\ (1)$

Also,



Area of quadrilateral ABCD = area of △ ABC + area of △ CDA

Since, $ar(\triangle \text{ ABC}) = ar(\triangle \text{ CDA})$

∴Area of the quadrilateral ABCD = 2 $ar(\triangle\text{ABC})$

$ar(\triangle \text{ ABC}) = \frac{1}{2}(\text{area of quadrilateral ABCD}) \ \dots\dots\dots\ (2)$

Using the equations (1) and (2)

$ar(\triangle \text{ ABD}) = ar(\triangle \text{ ABC})$

$\Rightarrow \frac{1}{2}\text{BD} = \frac{1}{2}\text{CL}$

⇒ CL = BD

⇒ DC|| AB

Similarly,

DC = AB and AD = BC

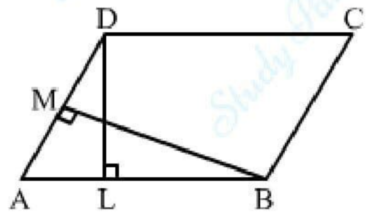
Hence, ABCD is a parallelogram

Area of parallelogram ABCD = base × height = 5 × 7= 35 cm^2

Solution 3:

Given, BM = 8cm, DL = 6cm and AB = 10cm

Area of the parallelogram ABCD = base ×height



$\Rightarrow AB \times DL = AD \times BM$

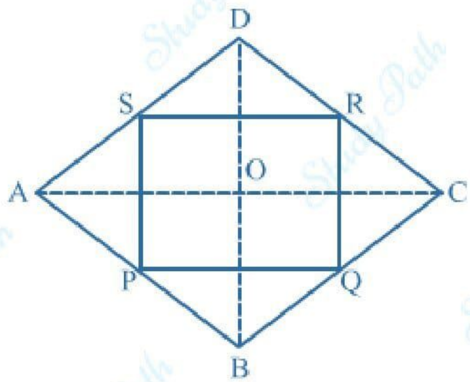
$\Rightarrow 10 \times 6 = AD \times 8$

$\Rightarrow AD \times 8 = 60$

$\Rightarrow AD = \frac{60}{8}$

$\Rightarrow AD = 7.5$ cm

Solution 4: Let, ABCD is a rhombus with P, Q, R and S as the midpoints of AB, BC, CD and DA.



AC and BD diagonals are joined.

In ΔABC,

$PQ = \frac{1}{2}AC$ [By using midpoint theorem]

$\Rightarrow PQ = \frac{1}{2} \times 16 = 8$ cm

In △DAC,

$SR = \frac{1}{2}AC$ [By using midpoint theorem]

$\Rightarrow SR = \frac{1}{2} \times 12 = 6$ cm

Area of the rectangle PQRS = length × breadth = 6 × 8 =48 cm^2

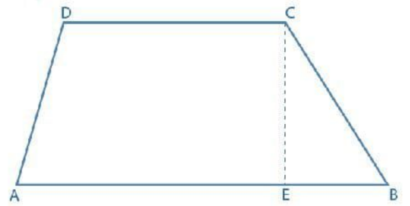
Solution 5: We know that,

Area of trapezium $= \frac{1}{2}$(sum of parallel sides $\times$ distance between them)

$= \frac{1}{2}(9+6) \times 8$

$= \frac{1}{2} \times 120$

$= 60$



Area of trapezium $= 60\ \text{cm}^2$

Solution 6:

(i)In $\triangle$BCD

Using the Pythagoras theorem, we have

$DB^2 + BC^2 = DC^2$

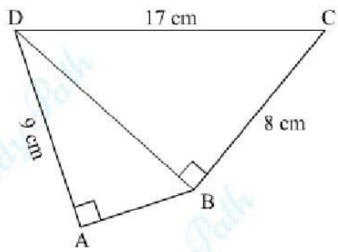
$\Rightarrow DB^2 + 8^2 = 17^2$

$\Rightarrow DB^2 = 289 - 64$

$\Rightarrow DB^2 = 225$

$\Rightarrow DB = \sqrt{225}$

$\Rightarrow DB = 15\text{cm}$



(i)

$\therefore$ Area of $\triangle$ BCD $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 8 \times 15 = 60\ \text{cm}^2$

In $\triangle$BAD,

Using the Pythagoras theorem, we can write

$DA^2 + AB^2 = DB^2$

$\Rightarrow AB^2 + 9^2 = 15^2$

$\Rightarrow AB^2 = 225 - 81$

$\Rightarrow AB^2 = 144$

$\Rightarrow AB = \sqrt{144}$

$\Rightarrow AB = 12\text{ cm}$

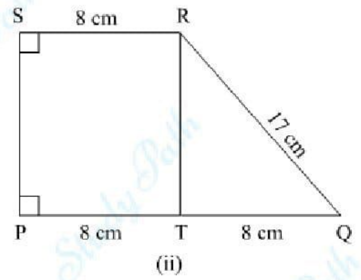
Area of $\triangle$ DAB $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 9 \times 12 = 54\ \text{cm}^2$

Area of quadrilateral ABCD = area of $\triangle$DAB + area of $\triangle$BCD

$= 54 + 60$

$= 114\ \text{cm}^2$

(ii) From the figure



In $\triangle$RTQ,

Using the Pythagoras theorem

RT2 + TQ2 = RQ2

$\Rightarrow RT^2 + 8^2 = 17^2$

$\Rightarrow RT^2 = 289 - 64$

$\Rightarrow RT^2 = 225$

$\Rightarrow RT = \sqrt{225}$

$\Rightarrow RT = 15\ \text{cm}$

Area of trapezium PQRS $= \frac{1}{2}$(sum of parallel sides $\times$ distance between them)

$= \frac{1}{2}(8 + 16) \times 15)$

$= 180\ \text{cm}^2$

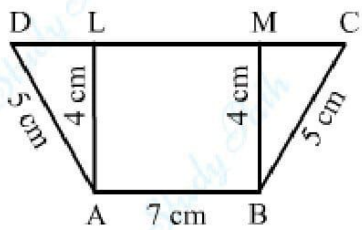
Therefore, the area of trapezium PQRS is $180\ \text{cm}^2$.

Solution 7:

Considering $\triangle$ALD

By using Pythagoras theorem, we can write

$AL^2 + DL^2 = AD^2$



$\Rightarrow DL^2 = 5^2 - 4^2$

$\Rightarrow DL^2 = 25 - 16$

$\Rightarrow DL^2 = 9$

$\Rightarrow DL = \sqrt{9}$

$\Rightarrow DL = 3\ \text{cm}$

Considering $\triangle$BMC

By using Pythagoras theorem, we can write

$MC^2 + MB^2 = CB^2$

$\Rightarrow MC = \sqrt{5^2 - 4^2} = \sqrt{25 - 16} = \sqrt{9} = 3$ cm

Now, from figure LM = AB = 7cm

$\therefore CD = DL + LM + MC = 3 + 7 + 3 = 13$ cm

Area of Trapezium ABCD $= \frac{1}{2}$(sum of parallel sides $\times$ distance between them)

$= \frac{1}{2} \times (CD + AB) \times AL$

$= \frac{1}{2} \times (13 + 7) \times 4$

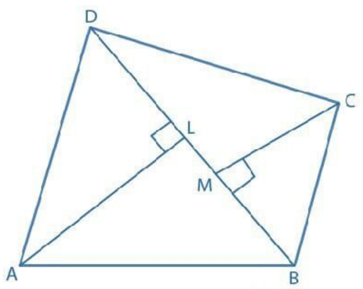
$= 40$

Area of Trapezium ABCD $= 40\ cm^2$

And length of DC = 13cm.

Solution 8:

From the figure, we can write



Area of $\triangle$ ABD $= \frac{1}{2} \times BD \times AL$

Area of $\triangle$ CBD $= \frac{1}{2} \times BD \times CM$

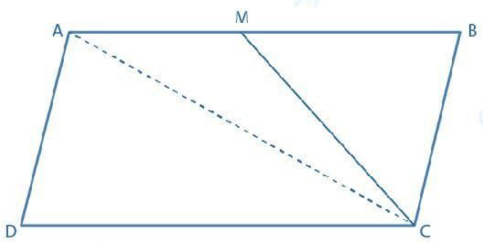
Area of quadrilateral ABCD = Area of $\triangle$ABD + Area of $\triangle$CBD

$= \frac{1}{2} \times BD \times AL + \frac{1}{2} \times BD \times CM$

$= \frac{1}{2} \times BD(AL + CM)$

Hence, proved

Solution 9: Join the diagonal AC



Here, the diagonal AC divides the parallelogram ABCD into two triangles having the same area

Area of $\triangle$ADC = Area of $\triangle$ABC ……. (1)

Since, $\triangle$ ADC and parallelogram ABCD are on the same base CD and between the same parallel lines DC and AM.

$$ar(\triangle \text{ADC}) = ar(\triangle \text{ ABC}) = \frac{1}{2}(\text{Area of parallelogram ABCD})$$

Also, M is the midpoint of AB

$$\therefore ar(\triangle \text{AMC}) = ar(\triangle \text{ BMC}) = \frac{1}{2} \times ar(\triangle \text{ABC}) = \frac{1}{2} \times ar(\triangle \text{ADC})$$

Area of AMCD = Area of $\triangle$ ADC + Area of $\triangle$ AMC

$$\Rightarrow 24 = ar(\triangle \text{ADC}) + \frac{1}{2} \times ar(\triangle \text{ADC})$$

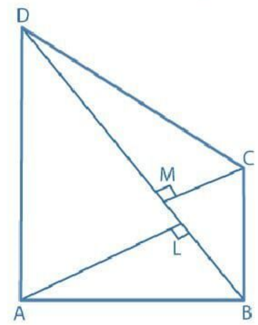
$$\Rightarrow 24 = \frac{3}{2} \times ar(\triangle ADC)$$

$$\text{Area of } \triangle \text{ADC} = \frac{24 \times 2}{3} = 16 \text{ cm}^2$$

From equation (1)

Area of $\triangle$ABC = 16 cm^2

Solution 10: We know that,



Area of quadrilateral ABCD = Area of $\triangle$ ABD + Area of $\triangle$ CBD

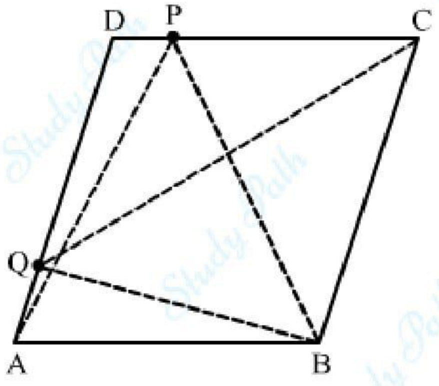
Now,

$$\text{Area of } \triangle \text{BAD} = \frac{1}{2} \times \text{BD} \times \text{AL} = \frac{1}{2} \times 14 \times 8 = 56 \text{ cm}^2$$

$$\text{Area of } \triangle \text{CBD} = \frac{1}{2} \times \text{BD} \times \text{CM} = \frac{1}{2} \times 14 \times 6 = 42 \text{ cm}^2$$

$\therefore$ Area of quadrilateral ABCD = 56 + 42 = 98 cm^2

Solution 11:



Here, $\triangle APB$ and parallelogram ABCD lie on the same base AB and between the same parallels AB and DC.

$\therefore$ Area of $\triangle APB = \frac{1}{2} \times ar(ABCD)$ ... (1)

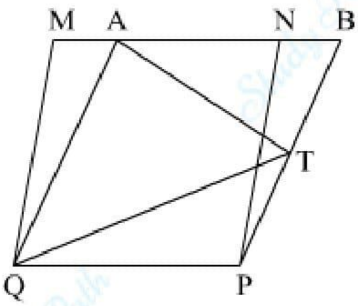
Also, $\triangle BQC$ and parallelogram ABCD lie on the same base BC and between the same parallels BC and AD

$\therefore$ Area of $\triangle BQC = \frac{1}{2}\, ar(ABCD)$ ... (2)

From equation (1) and (2), we get

Area of $\triangle APB$ = Area of $\triangle BQC$

Solution 12:



(i) Parallelograms MNPQ and ABPQ are on the same base PQ and lie between the same parallels PQ and MB.

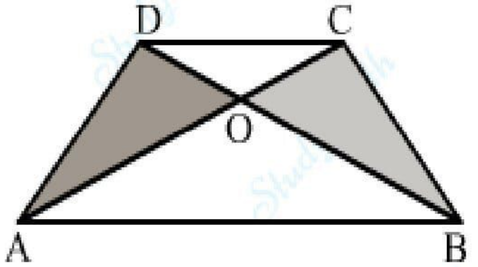
$\therefore$Area of parallelogram MNPQ = Area of parallelogram ABPQ ...(1)

(ii) $\triangle ATQ$ and parallelogram ABPQ lie on the same base AQ and between the same parallels AQ and BP.

$\therefore$ Area of $\triangle ATQ = \frac{1}{2} \times ar(ABPQ)$

$\Rightarrow$ Area of $\triangle ATQ = \frac{1}{2} \times ar(MNPQ)$ [From equation (1)]

Solution 13:



Here, $\triangle$AOD and $\triangle$DCB lie on the same base CD and between two parallel lines DC and AB.

We know that the triangles lying on the same base and parallels have equal area.

$\therefore$Area of $\triangle$CDA = Area of $\triangle$CDB ...(1)

Now,

$ar(\triangle \text{AOD}) = ar(\triangle \text{CDA}) - ar(\triangle \text{OCD})$ ... (2)

And

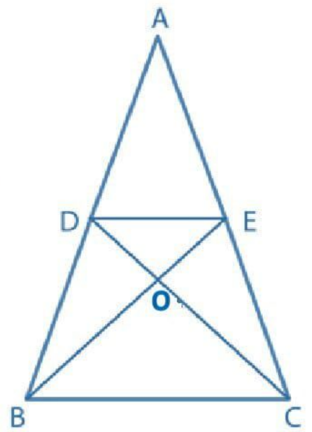
$ar(\triangle \text{BOC}) = ar(\triangle \text{CDB}) - ar(\triangle \text{OCD})$ from (1)

$\Rightarrow ar(\triangle \text{BOC}) = ar(\triangle \text{CDA}) - ar(\triangle \text{OCD})$ from(2)

$\Rightarrow ar(\triangle \text{BOC}) = ar(\text{AOD})$

$\therefore$ Area of $\triangle$AOD = Area of $\triangle$BOC

Solution:



(i) Here, $\triangle$DBE and $\triangle$DCE lie on the same base DE between the parallel lines BC and DE.

$\therefore$Area of $\triangle$ DBE = Area of $\triangle$ DCE ...(1)

By adding $\triangle$ ADE both sides

$ar(\triangle \text{DBE}) + ar(\triangle \text{ADE}) = ar(\triangle \text{DCE}) + ar(\triangle \text{ADE})$

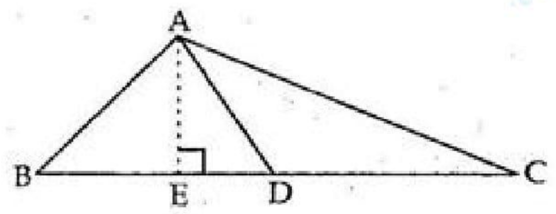
$ar(\triangle \text{ABE}) = ar(\triangle \text{ACD})$ **Proved**

(ii) Subtracting $ar(\triangle \text{ODE})$ from equation (1)

$ar(\triangle \text{DBE}) - ar(\triangle \text{ODE}) = ar(\triangle \text{DCE}) - ar(\triangle \text{ODE})$

$ar(\triangle \text{OBD}) = ar(\triangle \text{OCE})$

Solution 15: Let AD is a median of ΔABC and D is the midpoint of BC. AD divides ΔABC in two triangles: ΔABD and ΔADC.



In △ ABD

$ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{BD} \times \text{AE}$

Also, In ΔADC

$ar(\triangle \text{ADC}) = \frac{1}{2} \times \text{DC} \times \text{AE}$

Since, D is the median

$\therefore \text{BD} = \text{DC}$

$ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{BD} \times \text{AE}$

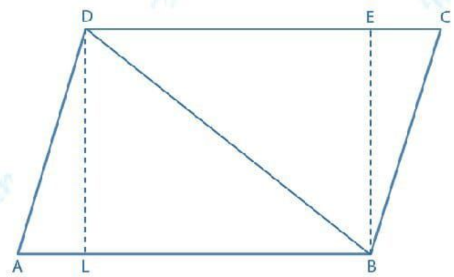
$\Rightarrow ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{DC} \times \text{AE}$

$ar(\triangle \text{ABD}) = ar(\triangle \text{ADC})$

Therefore, median divides a triangle into two triangles of equal area.

Solution 16:

Let ABCD be a parallelogram and BD be its diagonal.



To prove: $ar(\Delta\text{ABD}) = ar(\Delta\text{CBD})$

$ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{AB} \times \text{DL}$ ... (1)

$ar(\triangle \text{CBD}) = \frac{1}{2} \times \text{CD} \times \text{BE}$ ... (2)

Since, ABCD is a parallelogram.

It can be written as AB || CD and

AB = CD and DL = BE ...(3)

From equation (1) and (3), we get

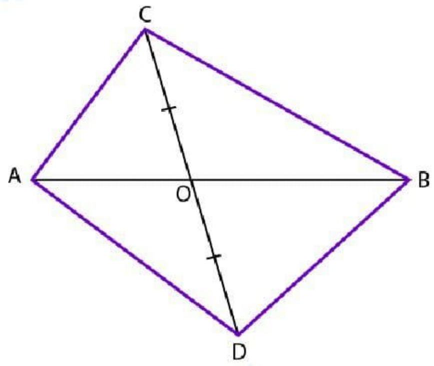
$ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{CD} \times \text{BE}$

$\Rightarrow ar(\triangle \text{ABD}) = ar(\Delta\text{CBD})$

Therefore, it is proved that a diagonal divides a parallelogram into two triangles of equal area.

Solution 17: From the figure we know that median of a triangle divides it into triangles of equal area

We know that AO is the median of $\triangle$ ACD and BO is the median of $\triangle$ BCD



$\therefore ar(\triangle COA) = ar(\triangle DOA)$ .... (1)

And $ar(\triangle COB) = ar(\triangle DOB)$ .... (2)

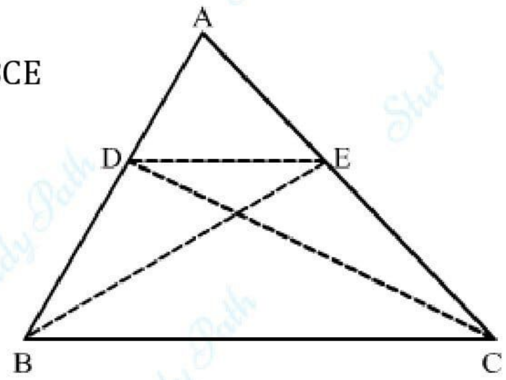
By adding equations (1) and (2), we get

$\therefore ar(\triangle COA) + ar(\triangle COB) = ar(\triangle DOA) + ar(\triangle DOB)$

$\Rightarrow ar(\triangle ABC) = ar(\triangle ABD)$

Therefore, it is proved that ar ($\triangle$ ABC) = ar ($\triangle$ ABD).

Solution 18: From the figure we know that $\triangle$ BCD and $\triangle$ BCE have equal area and lie on the same base BC.

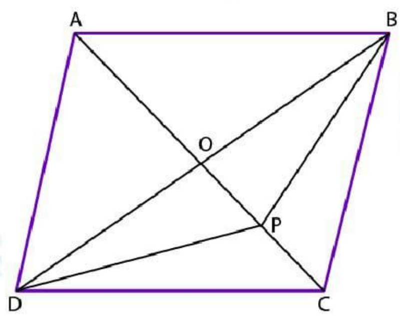


So, $\triangle$ BCD and $\triangle$ BCE lie between the same parallel lines

$\therefore$DE || BC

Therefore, it is proved that DE || BC.

Solution 19:



Join the diagonal BD.

From the figure we know that AC and BD are the diagonals intersecting at point O.

We know that the diagonals of a parallelogram bisect each other.

Thus, we get O as the midpoint of AC and BD.

Median of triangle divides it into two triangles having equal area.

In $\triangle$ABD, OA is the median

$\therefore ar(\triangle AOD) = ar(\triangle AOB)$ ...(1)

In $\triangle$BPD, OP is the median

$\therefore ar(\triangle OPD) = ar(\triangle OPB)$ ...(2)

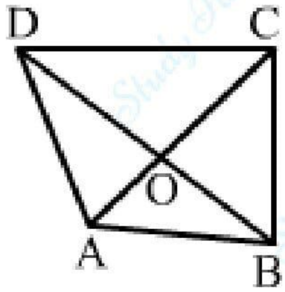
By adding equations (1) and (2), we get

$ar(\triangle AOD) + ar(\triangle OPD) = ar(\triangle AOB) + ar(\triangle OPB)$

$\therefore ar(\triangle ADP) = ar(\triangle ABP)$

Solution 20:

Given, BO = OD



From the figure we know that AO is the median of $\triangle$ ABD

$\therefore ar(\triangle AOD) = ar(\triangle AOB)$ ...(1)

Also, OC is the median of $\triangle$ CBD

$\therefore ar(\triangle DOC) = ar(\triangle BOC)$ ...(2)

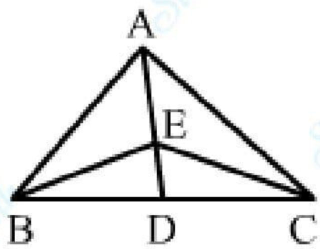
By adding equations (1) and (2, we get

$ar(\triangle AOD) + ar(\triangle DOC) = ar(\triangle AOB) + ar(\triangle BOC)$

$\Rightarrow ar(\triangle ADC) = ar(\triangle ABC)$

Therefore, it is proved that $ar(\triangle ABC) = ar(\triangle ADC)$.

Solution 21: From the figure we know that BE is the median of $\triangle$ ABD



$\therefore ar(\triangle BDE) = ar(\triangle ABE)$

$\Rightarrow ar(\triangle BDE) = \frac{1}{2} \times ar(\triangle ABD)$ ... (1)

Also, CE is the median of $\triangle$ ADC

$\therefore ar(\triangle CDE) = ar(\triangle ACE)$

$\Rightarrow ar(\triangle CDE) = \frac{1}{2} \times ar(\triangle ACD)$ ... (2)

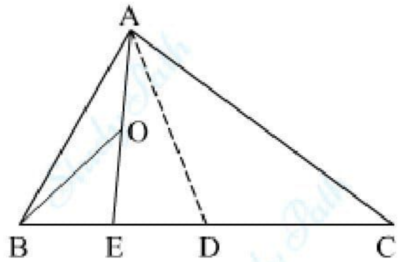
By adding equations (1) and (2), we get

$\text{ar}(\triangle BDE) + \text{ar}(\triangle CDE) = \frac{1}{2} \times ar(\triangle ABD) + \frac{1}{2} \times ar(\triangle ACD)$

$\Rightarrow ar(\triangle BEC) = \frac{1}{2} \times \left(ar(\triangle ABD) + ar(\triangle ACD)\right)$

$\Rightarrow ar(\triangle \mathrm{BEC}) = \frac{1}{2} ar(\triangle \mathrm{ABC})$

Solution 22:



O is the midpoint of AE.

We know that BO is the median of △ BAE

$ar(\triangle \mathrm{BOE}) = \frac{1}{2} \times \mathrm{ar}(\triangle \mathrm{ABE})$ ... (1)

E is the midpoint of BD

AE divides the △ ABD into two triangles of equal area

$\therefore ar(\triangle \mathrm{ABE}) = \frac{1}{2} \times ar(\triangle \mathrm{ABD})$ ... (2)

D is the midpoint of BC

$\therefore ar(\triangle \mathrm{ABD}) = \frac{1}{2} \times ar(\triangle \mathrm{ABC})$ ... (3)

Now, from equation (1), we get

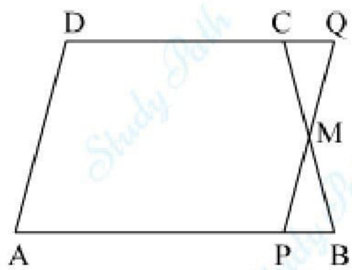
$ar(\triangle \mathrm{BOE}) = \frac{1}{2} \times \mathrm{ar}(\triangle \mathrm{ABE})$

$\Rightarrow ar(\triangle \mathrm{BOE}) = \frac{1}{2} \times \frac{1}{2} \times ar(\triangle \mathrm{ABD})$ From (2)

$\Rightarrow ar(\triangle \mathrm{BOE}) = \frac{1}{4} \times \frac{1}{2} \times ar(\triangle \mathrm{ABC})$ From (3)

$\Rightarrow ar(\triangle \mathrm{BOE}) = \frac{1}{8} ar(\triangle \mathrm{ABC})$

Solution 23:



In △MCQ and △MPB, ∠QCM and ∠PBM are alternate angles

$\therefore \angle \mathrm{QCM} = \angle \mathrm{PBM}$

$\mathrm{CM} = \mathrm{BM}$ [∵ M is the midpoint of BC]

$\angle \mathrm{CMQ} = \angle \mathrm{PBM}$ [Vertically opposite angles]

By ASA congruence criterion

$\triangle \mathrm{MCQ} \cong \triangle \mathrm{MPB}$

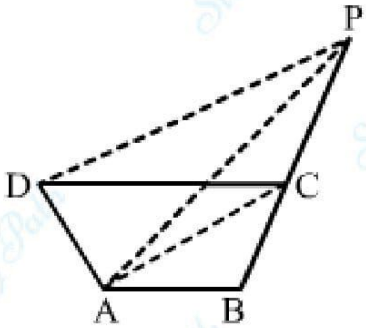
$\therefore ar(\triangle \mathrm{MCQ}) = ar(\triangle \mathrm{MPB})$

Now,

$ar(\text{ABCD}) = ar(\text{APQD}) + ar(\text{DMPB}) - ar(\triangle\text{MCQ})$

$\Rightarrow ar(\text{ABCD}) = ar(\text{APQD})$

Solution 24:



From the figure we know that $\triangle$ ACP and $\triangle$ ACD lie on the same base AC between parallel lines AC and DP

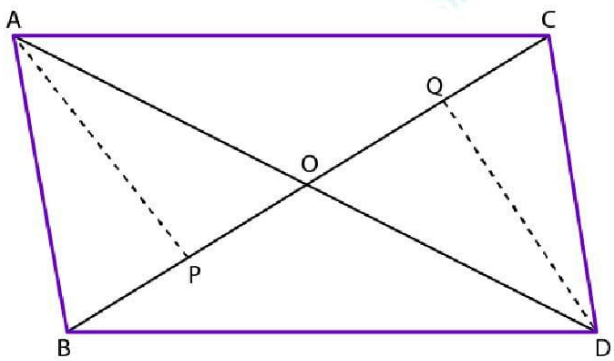
$\therefore ar(\triangle\text{ACP}) = ar(\triangle\text{ACD})$

$ar(\triangle\text{ACP}) + ar(\triangle\text{ABC}) = ar(\triangle\text{ACD}) + ar(\triangle\text{ABC})$

$ar(\triangle\text{ABP}) = ar(\text{ABCD})$

Solution 25:

Construct AP $\perp$ BC and DQ $\perp$ BC



Now,

$ar(\triangle\ \text{ABC}) = \frac{1}{2} \times \text{BC} \times \text{AP} \qquad \ldots(1)$

In the same way

$ar(\triangle\ \text{BCD}) = \frac{1}{2} \times \text{BC} \times \text{DQ} \qquad \ldots(2)$

Equating (1) and (2), we get

$\frac{1}{2} \times \text{BC} \times \text{AP} = \frac{1}{2} \times \text{BC} \times \text{DQ}$

$\Rightarrow \text{AP} = \text{DQ}$

In $\triangle$AOP and $\triangle$QOD

$\angle\text{APO} = \angle\text{DQO} \qquad$ [Each 90°]

$\angle AOP = \angle DOQ$ [Vertically opposite angles]

By AAS congruence criterion

$\triangle AOP \cong \triangle QOD$

$OA = OD$ [CPCT]

Therefore, it is proved that BC bisects AD.

Solution 26: In $\triangle ADM$ and $\triangle PCM$

$\angle ADM = \angle PCM$ [Alternate angles]

$AD = CP$ [$AD = BC = CP$]

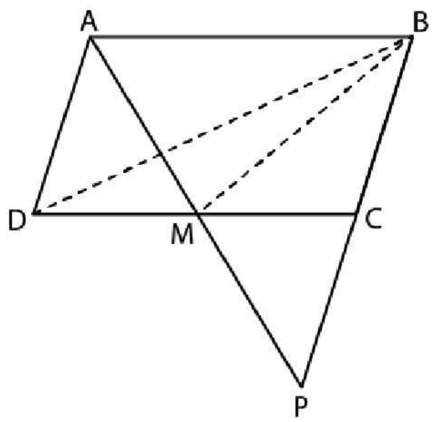
$\angle AMD = \angle PMC$ [Vertically opposite angles]

By ASA congruence criterion

$\triangle ADM \cong \triangle PCM$

$ar(\triangle ADM) = ar(\triangle PCM)$

$DM = CM$ [CPCT]



We know that BM is the median of $\triangle BDC$

$\therefore ar(\triangle DMB) = ar(\triangle CMB)$

$\Rightarrow ar(\triangle BDC) = 2\times ar(\triangle DMB)$

$\Rightarrow ar(\triangle BDC) = 2 \times 7 = 14 \text{ cm}^2$

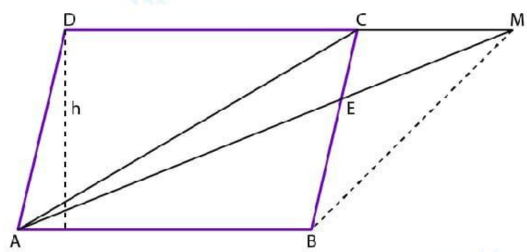
$ar(ABCD) = 2\times ar(\triangle BDC) = 2 \times 14 = 28 \text{ cm}^2$

Therefore, area of parallelogram ABCD is 28 cm^2.

Solution 27:

Construct lines AC and BM.

Let us take h as the distance between AB and CD



$ar(\triangle ACD) = \frac{1}{2} \times CD \times h$

$ar(\triangle ABM) = \frac{1}{2} \times AB \times h$

$ar(\triangle ABM) = \frac{1}{2} \times CD \times h$ [$AB = CD$]

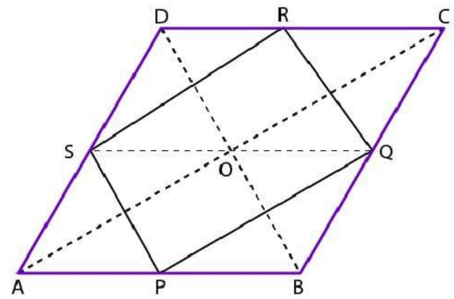
$ar(\triangle ABM) = ar(\triangle ACD)$

$ar(\triangle ABM) + ar(\triangle ACM) = ar(\triangle ACD) + ar(\triangle ACM)$ [By adding $\triangle$ ACM on both sides]

$ar(ABMC) = ar(\triangle ADM)$

Solution 28:

$ar(||\text{gm PQRS}) = \frac{1}{2} \times ar\ (||\text{gmABCD})$



Construct diagonals AC, BD and SQ

In $\triangle$ADC, S and R are the midpoints of AD and CD

Using the midpoint theorem

SR || AC

In $\triangle$ABC, P and Q are the midpoints of AB and BC

Using the midpoint theorem

PQ || AC

∴PQ || AC || SR

⇒PQ || SR

In the same way we get SP || RQ

In $\triangle$ABD, O is the midpoint of AC and S is the midpoint of AD

Using the midpoint theorem

OS || AB

In $\triangle$ ABC, OQ || AB

∴ SQ || AB

We know that ABQS is a parallelogram

$\therefore ar(\triangle \text{SPQ}) = ar(||\text{gm ABQS})$ ... (1)

In the same way we get

$ar(\triangle \text{SRQ}) = \frac{1}{2} \times ar(||gmSQCD)$ ... (2)

By adding equations (1) and (2), we get

$ar(\triangle \text{SPQ}) + ar(\triangle \text{SRQ}) = \frac{1}{2} \times \{ar\ (||gm\ \text{ABQS}) + ar(||gm\ \text{SQCD})\}$

$ar(||gm\ \text{PQRS}) = \frac{1}{2} \times ar(||gm\ \text{ABCD})$

Solution 29:

Draw a line EF.

We know that the line segment joining the midpoint of two sides of a triangle is parallel to the third side.

$\therefore$FE || BC

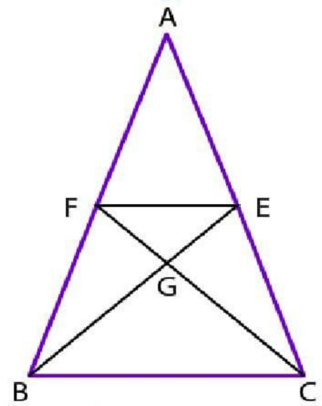
From the figure we know that $\triangle$BEF and $\triangle$CEF lie on the same base EF between the same parallel lines

$\therefore ar(\triangle \mathrm{BEF}) = ar(\triangle \mathrm{CEF})$

By subtracting $\triangle$ GEF both the sides

$ar(\triangle \mathrm{BEF}) - ar(\triangle \mathrm{GEF}) = ar(\triangle \mathrm{CEF}) - ar(\triangle \mathrm{GEF})$

$\therefore ar(\triangle \mathrm{BFG}) = ar(\triangle \mathrm{CEG})$ ...(1)



Median of a triangle divides it into two triangles having equal area

$ar(\triangle \mathrm{BEC}) = ar(\triangle \mathrm{ABE})$

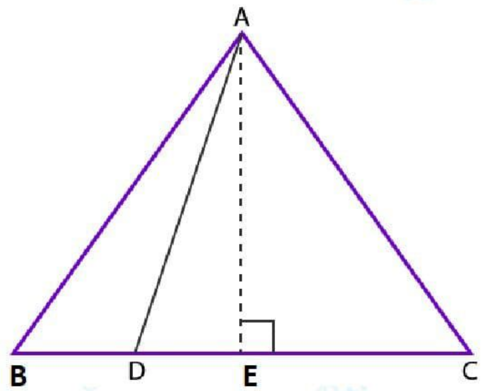
$\Rightarrow ar(\triangle \mathrm{BGC}) + ar(\triangle \mathrm{CEG}) = ar(||\mathrm{gm\ AFGE}) + ar(\triangle \mathrm{BFG})$

$\Rightarrow ar(\triangle \mathrm{BGC}) + ar(\triangle \mathrm{BFG}) = ar(\mathrm{AFGE}) + ar(\triangle \mathrm{BFG})$

$\Rightarrow ar(\triangle \mathrm{BGC}) = ar(\mathrm{AFGE})$

Solution 30:

Construct AE $\perp$ BC



We know that

$ar(\triangle \mathrm{ABD}) = \frac{1}{2} \times \mathrm{BD} \times \mathrm{AE}$ ... (1)

$ar(\triangle \mathrm{ABC}) = \frac{1}{2} \times \mathrm{BC} \times \mathrm{AE}$ ... (2)

$\because \mathrm{BD} = \frac{1}{2}\mathrm{DC}$ [Given]

Now,

$\mathrm{BC} = \mathrm{BD} + \mathrm{DC}$

$\Rightarrow \mathrm{BC} = \mathrm{BD} + 2\mathrm{BD}$

$\Rightarrow \mathrm{BC} = 3\mathrm{BD}$

$\Rightarrow \mathrm{BD} = \frac{1}{3}\mathrm{BC}$ ... (3)

Using equation (1) we get

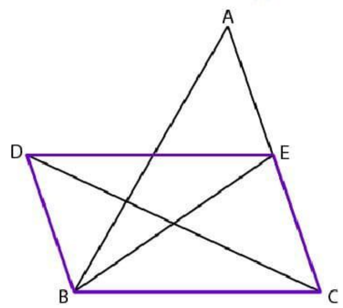
$ar(\triangle\,\mathrm{ABD}) = \frac{1}{2} \times \mathrm{BD} \times \mathrm{AE}$

$\Rightarrow ar(\triangle\,\mathrm{ABD}) = \frac{1}{2} \times \frac{1}{3} \times \mathrm{BC} \times \mathrm{AE}$

$\Rightarrow ar(\triangle\,\mathrm{ABD}) = \frac{1}{3} \times \frac{1}{2} \times \mathrm{BC} \times \mathrm{AE}$

$\Rightarrow ar(\triangle\,\mathrm{ABD}) = \frac{1}{3} \times ar(\triangle\,\mathrm{ABC})$ **Proved**

Solution 31:



Given: E is the midpoint of CA and $\mathrm{BD} = \frac{1}{2}\mathrm{CA}$

$\therefore \mathrm{BD} = \mathrm{CE}$

We know that BD || CA.

So, BD || CE and BD = CE

Hence, BCED is a parallelogram.

Now,

$\triangle$DBC and $\triangle$BEC lie on the same base BC and between the same parallel lines BC || DE

$\therefore ar(\triangle\,\mathrm{DBC}) = ar\,(\triangle\,\mathrm{BEC})$ ... (1)

Also, BE is the median of $\triangle$ ABC

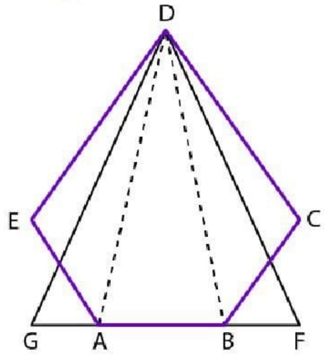
$\therefore\ ar(\triangle\,\mathrm{BEC}) = \frac{1}{2} \times ar(\triangle\,\mathrm{ABC})$

$\Rightarrow ar(\triangle\,\mathrm{DBC}) = \frac{1}{2} ar(\triangle\,\mathrm{ABC})$

$\Rightarrow ar(\triangle\,\mathrm{ABC}) = 2 \times ar(\triangle\,\mathrm{DBC})$

Therefore, it is proved that $ar(\triangle\,\mathrm{ABC}) = 2\ ar(\triangle\,\mathrm{DBC})$.

Solution 32:



$\triangle$DGA and $\triangle$AED have the same base AD and lie between the parallel lines AD and EG.

$ar(\triangle DGA) = ar(\triangle AED)$ ...(1)

$\triangle$DBC and $\triangle$BFD have the same base DB and lie between the parallel lines BD and CF.

$ar(\triangle DBF) = ar(\triangle BCD)$ ...(2)

By adding equations (1) and (2), we get

$ar(\triangle DGA) + ar(\triangle DBF) = ar(\triangle AED) + ar(\triangle BCD)$

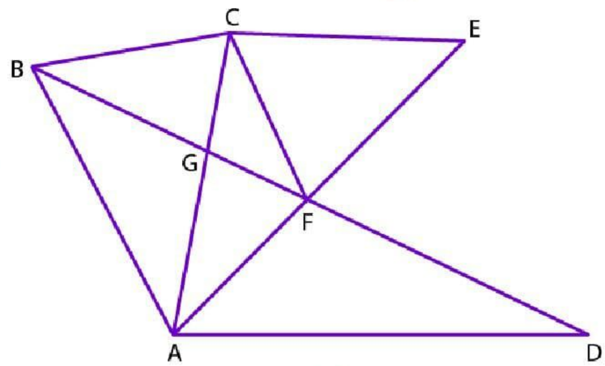
By adding $\triangle$ABD both sides

$ar(\triangle DGA) + ar(\triangle DBF) + ar(\triangle ABD) = ar(\triangle AED) + ar(\triangle BCD) + ar(\triangle ABD)$

$\Rightarrow ar(\triangle DGF) = ar(\text{pentagon ABCDE})$

Therefore, it is proved that $ar(\text{pentagon ABCDE}) = ar(\triangle \text{DGF})$.

Solution 33:



From the figure we know that $\triangle$ BCF and $\triangle$ ACF lie on the same base CF between the same parallels CF and BA

$ar(\triangle BCF) = ar(\triangle ACF)$

By subtracting $\triangle$CGF both sides

$ar(\triangle BCF) - ar(\triangle CGF) = ar(\triangle ACF) - ar(\triangle CGF)$

$ar(\triangle CBG) = ar(\triangle AFG)$

Therefore, it is proved that $ar(\triangle \text{CBG}) = ar(\triangle \text{AFG})$.

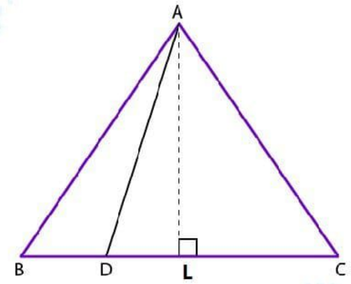
Solution 34:

$ar(\triangle \text{ABD}) = \frac{1}{2} \times \text{BD} \times \text{AL}$

$ar(\triangle \mathrm{ADC}) = \frac{1}{2} \times \mathrm{DC} \times \mathrm{AL}$

Given: $\mathrm{BD} : \mathrm{DC} = \mathrm{m} : \mathrm{n}$

$\Rightarrow \mathrm{BD} = \mathrm{DC} \times \frac{m}{n}$



$ar(\triangle \mathrm{ABD}) = \frac{1}{2} \times \mathrm{BD} \times \mathrm{AL}$

$\Rightarrow ar(\triangle \mathrm{ABD}) = \frac{1}{2} \times \left(\mathrm{DC} \times \frac{m}{n}\right) \times \mathrm{AL}$

$\Rightarrow ar(\triangle \mathrm{ABD}) = \frac{m}{n} \times \frac{1}{2} \times \mathrm{DC} \times \mathrm{AL}$

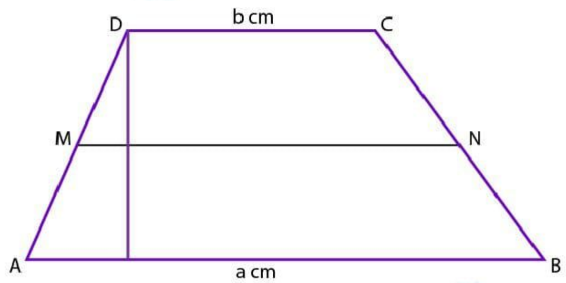
$\Rightarrow ar(\triangle \mathrm{ABD}) = \frac{m}{n} \times ar(\Delta ADC)$

$\Rightarrow \frac{ar(\triangle \mathrm{ABD})}{ar(\triangle \mathrm{ADC})} = \frac{m}{n}$

$\Rightarrow ar(\triangle \mathrm{ABD}) : ar(\triangle \mathrm{ADC}) = m : n$

Therefore, it is proved that $ar(\triangle \mathrm{ABD}) : ar(\triangle \mathrm{ADC}) = m : n$.

Solution 35:



Construction: Draw a perpendicular from point D to the opposite side AB, meeting AB at Q and MN at P.

Let length $\mathrm{DQ} = h$

Given: M and N are the midpoints of AD and BC respectively.

So, $\mathrm{MN} \parallel \mathrm{AB} \parallel \mathrm{DC}$ and $\mathrm{MN} = \frac{1}{2}(\mathrm{AB} + \mathrm{DC}) = \frac{1}{2}(\mathrm{a} + \mathrm{b})$

M is the mid-point of AD and MP || AB so by converse of mid-point theorem,

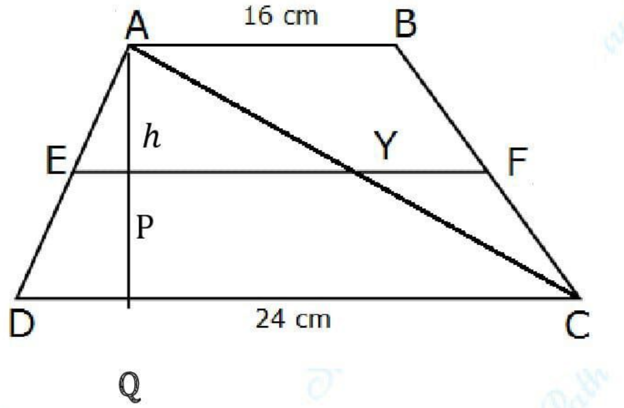
MP || AQ and P will be the mid-point of DQ.

$ar(\mathrm{DCNM}) = \frac{1}{2} \times \mathrm{DP}(\mathrm{DC} + \mathrm{MN}) = \frac{1}{2} h \left(b + \frac{a+b}{2}\right) = \frac{h}{4}(a + 3b)$

$ar(\text{MNBA}) = \frac{1}{2} \times \text{PQ}(\text{AB} + \text{MN}) = \frac{1}{2}h\left(a + \frac{a+b}{2}\right) = \frac{h}{4}(b + 3a)$

$ar(\text{DCNM}) : ar(\text{MNBA}) = (a + 3b) : (3a + b)$

Solution 36: Construction: Draw a perpendicular from point D to the opposite side CD, meeting CD at Q and EF at P.



Then, P is the mid-point of AQ.

i.e., $\text{AP} = \text{PQ} = h$ (say)

Given: E and F are the midpoints of AD and BC respectively.

So, EF || AB || DC and $\text{EF} = \frac{1}{2}(\text{AB} + \text{CD})$

E is the mid-point of AD and EP || AB so by converse of mid -point theorem,

EP || DQ and P will be the mid-point of AQ.

$ar(\text{ABFE}) = \frac{1}{2} \times \text{AP}(\text{AB} + \text{EF})$

$= \frac{1}{2}h\left(AB + \frac{1}{2}(\text{AB} + \text{CD})\right)$

$= \frac{1}{2}h\left(\frac{2AB + AB + CD}{2}\right)$

$= \frac{h}{4}(3AB + CD)$

$= \frac{h}{4}(3 \times 16 + 24)$

$= \frac{h}{4}(48 + 24)$

$$= \frac{h}{4} \times 72$$

$$= 18h$$

$$ar(\text{EFCD}) = \frac{1}{2} \times \text{PQ}(\text{CD} + \text{EF})$$

$$= \frac{1}{2}h\left(CD + \frac{1}{2}(\text{AB} + \text{CD})\right)$$

$$= \frac{1}{2}h\left(\frac{2CD + AB + CD}{2}\right)$$

$$= \frac{h}{4}(AB + 3CD)$$

$$= \frac{h}{4}(16 + 3 \times 24)$$

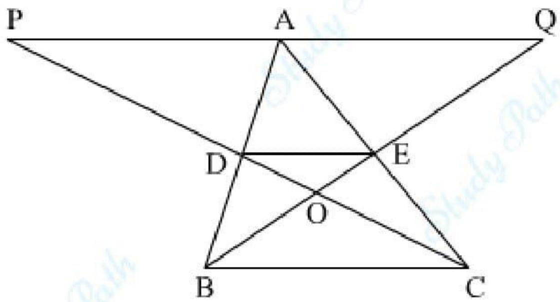
$$= \frac{h}{4}(16 + 72)$$

$$= \frac{h}{4} \times 88$$

$$= 22h$$

$$\text{So,} \frac{ar(\text{ABEF})}{ar(\text{EFCD})} = \frac{18h}{22h} = \frac{18}{22} = \frac{9}{11}$$

Solution 37:



In △△PAC,

PA || DE and E is the midpoint of AC

So, D is the midpoint of PC by converse of midpoint theorem.

$$\text{Also, DE} = \frac{1}{2}\text{PA} \qquad \ldots(1)$$

Similarly,

$$\text{DE} = \frac{1}{2}\text{AQ} \qquad \ldots(2)$$

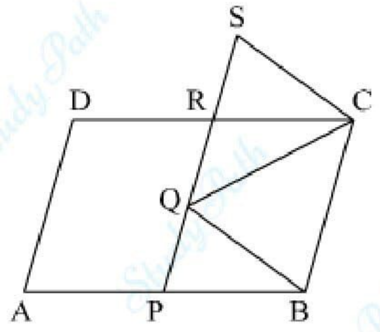
From (1) and (2) we have

PA = AQ

ΔABQ and ΔACP are on same base PQ and between same parallels PQ and BC

$\therefore ar(\Delta ABQ) = ar(\Delta ACP)$

Solution 38:



In ΔRSC and ΔPQB

$\angle CRS = \angle BPQ$ [CD || AB) so, corresponding angles are equal]

$\angle CSR = \angle BQP$ [SC || QB so, corresponding angles are equal]

SC = QB [BQSC is a parallelogram]

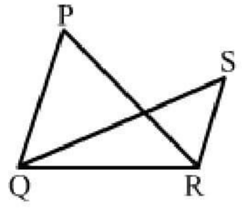
By AAS congruence criterion

$\Delta RSC \cong \Delta PQB$

$\therefore ar(\Delta RSC) = ar(\Delta PQB)$

Multiple Choice Questions (MCQ)

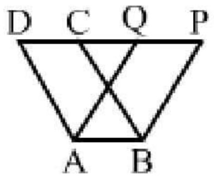
Solution 1: Correct option (b)



(b)

In this figure, both the triangles are on the same base (QR) but not on the same parallels.

Solution 2: Correct option (c)



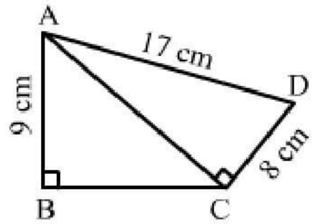
(c)

In this figure, the following polygons lie on the same base and between the same parallel lines:
(a) Parallelogram ABCD
(b) Parallelogram ABPQ

Solution 3: (a) triangles of equal areas

Solution 4: (c) 114 cm^2

Area of quadrilateral ABCD $= ar(\Delta ABC) + ar(\Delta ACD)$
In ΔACD



$\Rightarrow AD^2 = AC^2 + CD^2$
$\Rightarrow 17^2 = AC^2 + 8^2$
$\Rightarrow AC = \sqrt{17^2 - 8^2}$
$\Rightarrow AC = \sqrt{(17+8)(17-8)}$
$\Rightarrow AC = \sqrt{25 \times 9}$
$\Rightarrow AC = \sqrt{225} = 15 \text{ cm}$
$ar(\Delta ACD) = \frac{1}{2} \times 15 \times 8 = 60 \text{ cm}^2$

In ΔABC

$AC^2 = AB^2 + BC^2$

$\Rightarrow 15^2 = 9^2 + BC^2$

$\Rightarrow 225 = 81 + BC^2$

$\Rightarrow 255 - 81 = BC^2$

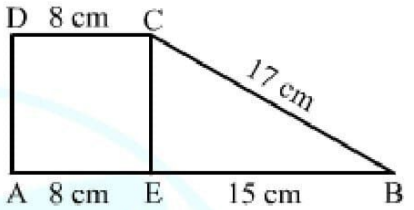
$\Rightarrow BC = \sqrt{144}$

$\Rightarrow BC = 12 \text{ cm}$

$ar(\Delta ABC) = \frac{1}{2} \times 12 \times 9 = 54 \text{ cm}^2$

Area of quadrilateral ABCD $= ar(\Delta ABC) + ar(\Delta ACD) = 54 \text{ cm}^2 + 60 \text{ cm}^2 = 114 \text{ cm}^2$

Solution 5: (c) 124 cm^2



In ΔBEC,

$BC^2 = CE^2 + EB^3$

$\Rightarrow 17^2 = CE^2 + 15^2$

$\Rightarrow CE = \sqrt{17^2 - 15^2}$

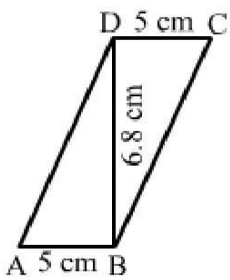
$\Rightarrow CE = \sqrt{289 - 225}$

$\Rightarrow CE = \sqrt{64} = 8 \text{ cm}$

Area of trapezium ABCD $= \frac{1}{2} \times$ (sum of parallel sides) $\times$ (distance between them)

$= \frac{1}{2} \times 31 \times 8 = 124 \text{ cm}^2$

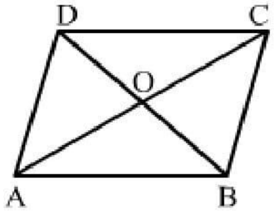
Solution 6: (c) 34 cm^2



$ar(\text{parallelogram ABCD}) = \text{base} \times \text{height} = 5 \times 6.8 = 34 \text{ cm}^2$

Solution 7: (d) 13 cm^2

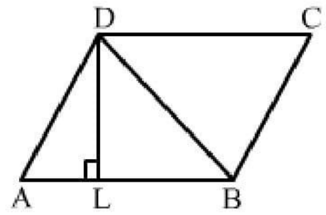
The diagonals of a parallelogram divides it into four triangles of equal areas.



$\therefore\ ar(\Delta OAB) = \frac{1}{4} \times ar(||gm\ ABCD)$

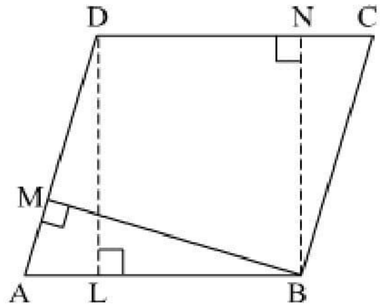
$\Rightarrow ar(\Delta OAB) = \frac{1}{4} \times 52 = 13 \text{ cm}^2$

Solution 8: (a) 40 cm^2



$ar(||\text{gm ABCD}) = \text{base} \times \text{height} = 10 \times 4 = 40\ cm^2$

Solution 9: (c) DC × DL

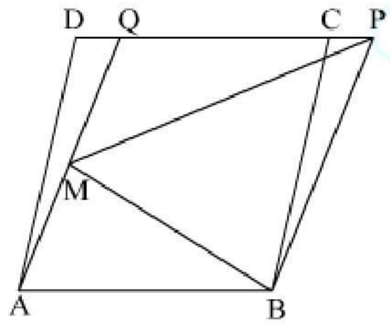


Area of a parallelogram is base into height.
Height = DL = NB
Base = AB = CD
So, area of parallelogram ABCD = DC × DL

Solution 10: (b) 1: 1

Parallelograms on the same base and between the same parallels are equal in area.
So, the ratio of their areas will be 1: 1.

Solution 11: (a) true



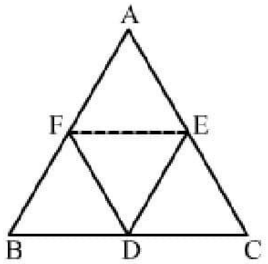
We know parallelogram on the same base and between the same parallels are equal in area.
Here, AB is the common base and AB || PD
Hence, $ar(\text{ABCD}) = ar(\text{ABPQ})$ ...(1)
Here, for the ΔBMP and parallelogram ABPQ, BP is the common base and they are between the common parallels BP and AQ
$\therefore ar(\Delta\text{BMP}) = \frac{1}{2} \times ar(||gm\ \text{ABPQ})$ ... (2)

From (1) and (2) we have

$ar(\Delta \text{BMP}) = \frac{1}{2} \times ar(||gm\ \text{ABCD})$

Thus, the given statement is true.

Solution 12: (a) $\frac{1}{2} \times ar(\Delta \text{ABC})$



D, E and F are the midpoints of sides BC, AC and AB respectively.

On joining FE, we divide $\triangle$ABC into 4 triangles of equal area.

Also, median of a triangle divides it into two triangles with equal area

$$ar(\text{AFDE}) = ar(\triangle\ \text{AFE}) + ar(\triangle\ \text{FED})$$
$$= 2ar(\triangle\ \text{AFE})$$
$$= 2 \times \frac{1}{4} ar(\triangle\ \text{ABC})$$
$$= \frac{1}{2} ar(\triangle\ \text{ABC})$$

Hence, the correct answer is option (a).

Solution 13: (b) 96 cm^2

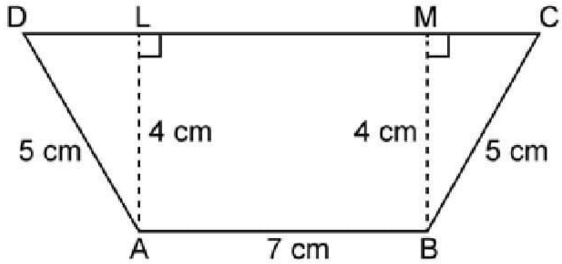
Area of the rhombus $= \frac{1}{2} \times$ product of diagonals $= \frac{1}{2} \times 12 \times 16 = 96\ \text{cm}^2$

Solution 14: (c) 65 cm^2

Area of the trapezium $= \frac{1}{2} \times$ (sum of parallel sides) $\times$ distance between them

$$= \frac{1}{2} \times (12 + 8) \times 6.5$$
$$= 65\ \text{cm}^2$$

Solution 15: (b) 40 cm^2



In right angled triangle MBC, we have:

$MC = \sqrt{5^2 - 4^2} = \sqrt{9} = 3$ cm

In right angled triangle ADL, we have:

$DL = \sqrt{5^2 - 4^2} = \sqrt{9} = 3$ cm

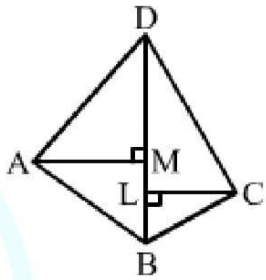
Now, $CD = ML + MC + LD = 7 + 3 + 3 = 13$ cm

$\therefore$ Area of the trapezium $= \frac{1}{2} \times$ (sum of parallel sides) $\times$ distance between them

$= \frac{1}{2} \times (13 + 7) \times 4$

$= 40\ cm^2$

Solution 16: (b) $128\ cm^2$



$ar(\text{quad ABCD}) = ar\ (\Delta\ ABD) + ar\ (\Delta\ DBC)$

$ar(\Delta ABD) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 16 \times 9 = 72\ cm^2$

$ar(\Delta DBC) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 16 \times 7 = 56\ cm^2$

$\therefore ar(quad\ ABCD) = 72 + 56 = 128\ cm^2$

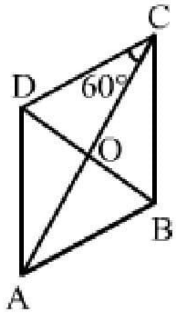
Solution 17: (a) $\sqrt{3} : 1$

ABCD is a rhombus. So, all of its sides are equal.

Now, $BC = DC$

$\Rightarrow \angle BDC = \angle DBC = x°$ [Angles opposite to equal sides are equal]

Also, $\angle BCD = 60°$



$\therefore\ x° + x° + 60° = 180°$

$\Rightarrow 2x° = 120°$

$\Rightarrow\ x° = 60°$

i. e., $\angle BCD = \angle BDC = \angle DBC = 60°$

So, ΔBCD is an equilateral triangle.

$\therefore BD = BC = a$

Also, $\text{OB} = \dfrac{a}{2}$

Now, in ΔOAB, we have:

$AB^2 = OA^2 + OB^2$

$\Rightarrow \text{OA}^2 = \text{AB}^2 - \text{OB}^2$

$\Rightarrow \text{OA}^2 = a^2 - \left(\dfrac{a}{2}\right)^2$

$\Rightarrow \text{OA}^2 = a^2 - \dfrac{a^2}{4}$

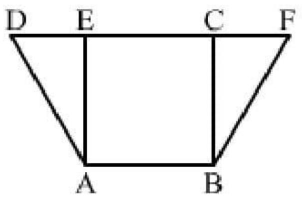
$\Rightarrow \text{OA}^2 = \dfrac{3a^2}{4}$

$\Rightarrow \text{OA} = \dfrac{\sqrt{3}a}{2}$

Now, $\text{AC} = 2 \times \text{OA} = 2 \times \dfrac{\sqrt{3}a}{2} = \sqrt{3}a$

$\therefore \text{AC} : \text{BD} = \sqrt{3}a : a = 3 : 1$

Solution 18: (d) 8 cm^2

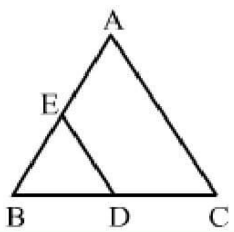


Since ||gm ABCD and ||gm ABFE are on the same base and between the same parallel lines, we have:

ar(||gm ABFE) = ar(||gm ABCD) = 25 cm^2

$\Rightarrow$ ar(ΔBCF) = ar(||gm ABFE) $-ar$(quad EABC) = (25 $-$17) = 8 cm^2

Solution 19: (b) 1: 4



ΔABC and ΔBDE are two equilateral triangles and D is the midpoint of BC.

Let $\mathrm{AB} = \mathrm{BC} = \mathrm{AC} = a$

Then, $\mathrm{BD} = \mathrm{BE} = \mathrm{ED} = \dfrac{a}{2}$

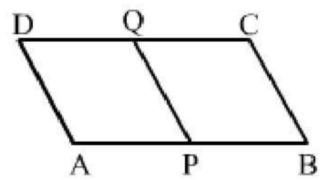
$$\therefore \frac{ar(\Delta \mathrm{BDE})}{ar(\Delta \mathrm{ABC})} = \frac{\frac{\sqrt{3}}{4}\mathrm{AB}^2}{\frac{\sqrt{3}}{4}\ \mathrm{BE}^2} = \frac{\left(\frac{a}{2}\right)^2}{a^2} = \frac{1}{4}$$

So, required ratio $=1:4$

Solution 20: (a) 8 cm^2

Let the distance between AB and CD be h cm.

Then $ar(||\mathrm{gm\ APQD}) = \mathrm{AP} \times \mathrm{h}$

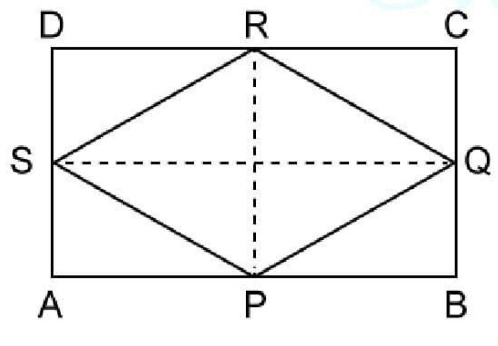


$= \dfrac{1}{2} \times \mathrm{AB} \times h \qquad [AP = 12AB]$

$= \dfrac{1}{2} \times ar(||gm\ \mathrm{ABCD}) \qquad [\ ar(||\ gm\ \mathrm{ABCD}) = \mathrm{AB} \times h\ )$

$\therefore\ ar\ (||gm\ \mathrm{APQD}) = \dfrac{1}{2} \times 16 = 8\ \mathrm{cm}^2$

Solution 21: (d) rhombus of 24 cm^2



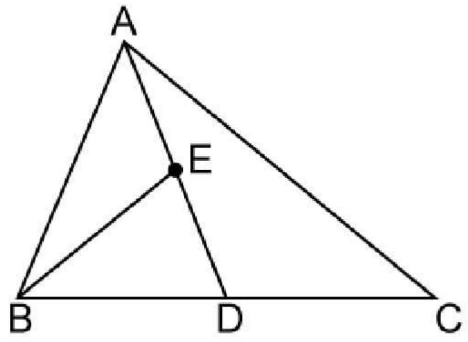
We know that the figure formed by joining the midpoints of the adjacent sides of a rectangle is a rhombus.

So, PQRS is a rhombus and SQ and PR are its diagonals.

i.e., $\mathrm{SQ} = 8$ cm and $\mathrm{PR} = 6$ cm

$\therefore\ ar(\text{rhombus PQRS}) = \dfrac{1}{2} \times$ product of diagonals $= \dfrac{1}{2} \times\ 8\ \times 6 = 24\ \mathrm{cm}^2$

Solution 22: (c) $\frac{1}{4}\ ar\ (\Delta\ \mathrm{ABC}\ )$



Since D is the mid-point of BC, AD is a median of ΔABC and BE is the median of ΔABD.

We know that a median of a triangle divides it into two triangles of equal areas.

$$\text{i.e.,}\ ar(\Delta\mathrm{ABD}) = \frac{1}{2}\ ar(\Delta\mathrm{ABC}) \qquad \ldots\text{(i)}$$

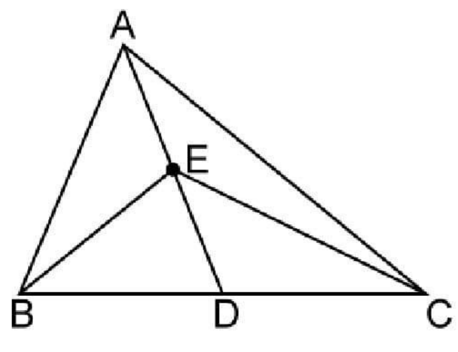
$$\Rightarrow\ ar(\Delta\mathrm{BED}) = \frac{1}{2}\ ar(\Delta\mathrm{ABD}) \qquad \ldots\text{(ii)}$$

From (i) and (ii), we have:

$$ar(\Delta\mathrm{BED}) = \frac{1}{2}\times\frac{1}{2}\times ar(\Delta\mathrm{ABC})$$

$$\therefore\ ar(\Delta\mathrm{BED}) = \frac{1}{4}\times ar(\Delta\mathrm{ABC})$$

Solution 23: (a) $\frac{1}{2}\ ar(\Delta\mathrm{ABC})$



Since E is the midpoint of AD, BE is a median of ΔABD.

We know that a median of a triangle divides it into two triangles of equal areas.

$$\text{i.e.,}\ ar(\Delta\mathrm{BED}) = \frac{1}{2}\times ar(\Delta\mathrm{ABD}) \qquad \ldots\text{(i)}$$

Since E is the midpoint of AD, CE is a median of ΔADC.

We know that a median of a triangle divides it into two triangles of equal areas.

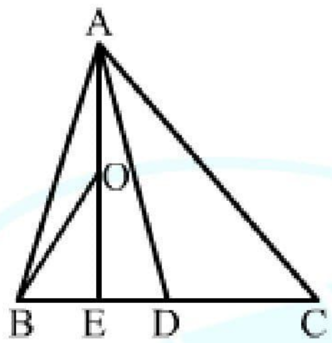
i. e. , $ar(\Delta CED) = \frac{1}{2} \times ar(\Delta ADC)$ ... (ii)

Adding (i) and (ii), we have:

$ar(\Delta BED) + ar(\Delta CED) = \frac{1}{2} \times ar(\Delta ABD) + \frac{1}{2} \times ar(\Delta ADC)$

$\Rightarrow ar(\Delta BEC) = \frac{1}{2}(\Delta ABD + \Delta ADC) = \frac{1}{2}\Delta ABC$

Solution 24: (d) $\frac{1}{8} \times ar\ (\Delta\ ABC)$



Given: D is the midpoint of BC, E is the midpoint of BD and O is the mid-point of AE.

Since D is the midpoint of BC, AD is the median of ΔABC.

E is the midpoint of BC, so AE is the median of ΔABD. O is the midpoint of AE, so BO is median of ΔABE.

We know that a median of a triangle divides it into two triangles of equal areas.

$i.e., ar(\Delta ABD) = \frac{1}{2} \times ar(\Delta ABC)$ ...(i)

$ar(\Delta ABE) = \frac{1}{2} \times ar(\Delta ABD)$ ... (ii)

$ar(\Delta BOE) = \frac{1}{2} \times ar(\Delta ABE)$ ... (iii)

From (i), (ii) and (iii), we have:

$ar(\Delta BOE) = \frac{1}{2} \times ar(\Delta ABE)$

$ar(\Delta BOE) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} ar(\Delta ABC)$

$\therefore ar(\Delta BOE) = \frac{1}{8} ar(\Delta ABC)$

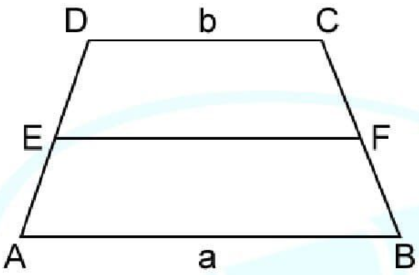
Solution 25: (a) 1:2

If a triangle and a parallelogram are on the same base and between the same parallels, then the area of the triangles is half the area of the parallelogram.

i. e., area of triangle $= \frac{1}{2} \times$ area of parallelogram

$\therefore$ Required ratio = area of triangle : area of parallelogram $= \frac{1}{2} : 1 = 1 : 2$

Solution 26: (c) $(3a + b) : (a + 3b)$



Clearly, $EF = \frac{1}{2}(a + b)$ [Mid point theorem]

Let d be the distance between AB and EF.

Then d is the distance between DC and EF.

Now, we have:

$$ar(trapezium\ ABEF) = \frac{1}{2}\left(a + \frac{a+b}{2}\right)d = \frac{(3a+b)d}{4}$$

$$ar(trapezium\ EFCD) = \frac{1}{2}\left(b + \frac{a+b}{2}\right)d = \frac{(a+3b)d}{4}$$

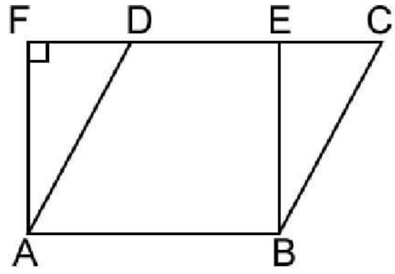
$$\therefore \text{ Required ratio} = \frac{(3a+b)d}{4} : \frac{(a+3b)d}{4} = (3a + b) : (a + 3b)$$

Solution 27: (d) all of these

In all the mentioned quadrilaterals, a diagonal divide them into two triangles of equal areas.

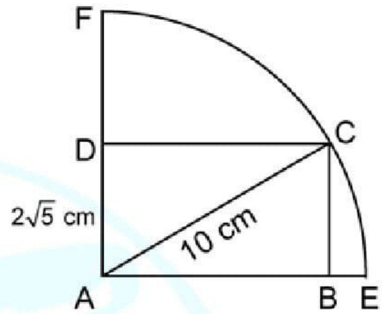
Solution 28: (c) perimeter of ABCD > perimeter of ABEF

Parallelogram ABCD and rectangle ABEF lie on the same base AB, i.e., one side is common in both the figures.



In ||gm ABCD, we have:
AD is the hypotenuse of right-angled triangle ADF.
So, AD > AF
∴ Perimeter of ABCD > perimeter of ABEF

Solution 29: (b) 40 cm²



Radius of the circle, AC = 10 cm
Diagonal of the rectangle, AC = 10 cm
Now,
$AB^2 = AC^2 - BC^2$
$\Rightarrow AB = \sqrt{AC^2 - BC^2}$
$\Rightarrow AB = \sqrt{10^2 - (2\sqrt{5})^2}$
$\Rightarrow AB = \sqrt{100 - 20}$
$\Rightarrow AB = \sqrt{80} = 4\sqrt{5}$ cm
∴ Area of the rectangle $= AB \times AD = 2\sqrt{5} \times 4\sqrt{5} = 40\text{ cm}^2$

Solution 30: (d) In a trapezium ABCD, it is given that AB || DC and the diagonals AC and BD intersect at O. Then $ar(\Delta AOB) = ar(\Delta COD)$.

Consider ΔADB and ΔADC, which do not lie on the same base but lie between same parallel lines.
i.e., $ar(\Delta ADB) \neq ar(\Delta ADC)$
Subtracting $ar(\Delta AOD)$ from both sides, we get:
$ar(\Delta ADB) - ar(\Delta AOD) \neq ar(\Delta ADC) - ar(\Delta AOD)$
$\Rightarrow ar(\Delta\, AOB) \neq ar(\Delta\, COD)$

Solution 31: (b) Area of a || gm $= \frac{1}{2} \times$ base × corresponding height

Solution 32: (c) I and II

Statement I is true, because if a parallelogram and a rectangle lie on the same base and between the same parallel lines, then they have the same altitude and therefore equal areas.

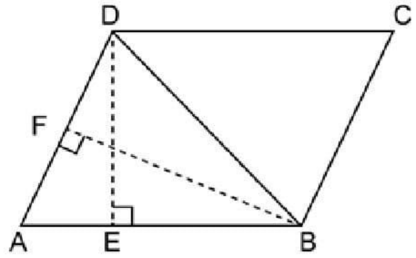
Statement II is also true as area of a parallelogram = base × height

$\Rightarrow AB \times DE = AD \times BF$

$\Rightarrow 10 \times 6 = 8 \times AD$

$\Rightarrow \ AD \ = \dfrac{60}{8} \ = \ 7.5 \text{ cm}$

Hence, statements I and II are true.



Solution 33: (a) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

In trapezium ABCD, ΔABC and ΔABD are on the same base and between the same parallel lines.

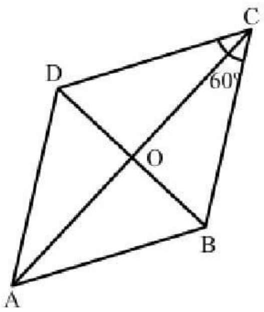
$\therefore ar(\Delta ABC) = ar(\Delta ABD)$

$\Rightarrow ar(\Delta ABC) - ar(\Delta AOB) = ar(\Delta ABD) - ar(\Delta AOB)$

$\Rightarrow \text{ar}(\Delta BOC) = \text{ar}(\Delta AOD)$

∴ Assertion (A) is true and, clearly, reason (R) gives (A).

Solution 34: (b) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.



Reason (R) is clearly true.

The explanation of assertion (A) is as follows:

ABCD is a rhombus. So, all of its sides are equal.

Now, BC = DC

$\Rightarrow \angle BDC = \angle DBC = x°$

Also, $\angle BCD = 60°$

$\therefore x° + x° + 60° = 180°$

$\Rightarrow 2x° = 120°$

$\Rightarrow x° = 60°$

$\therefore \angle BCD = \angle BDC = \angle DBC = 60°$

So, ΔBCD is an equilateral triangle.

i.e., BD = BC = a

$\therefore \ OB = \frac{a}{2}$

Now, in Δ OAB, we have:

$OA^2 = AB^2 - OB^2 = a^2 - \left(\frac{a}{2}\right)^2 = \frac{3a^2}{4}$

$\Rightarrow OA = \frac{\sqrt{3}a}{2}$

$\Rightarrow \ AC = 2 \times \frac{\sqrt{3}a}{2} = \sqrt{3}a$

$\therefore AC{:}BD = \sqrt{3}a : a = \sqrt{3} : 1$

Thus, assertion (A) is also true, but reason (R) does not give (A).

Hence, the correct answer is (b).

Solution 35: (a) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

Solution 36: (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

Explanation:

Reason (R):

$\therefore ar(\Delta ABC) = \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3}\ \text{cm}^2$

Thus, reason (R) is true.

Assertion (A):

Area of trapezium $= \frac{1}{2} \times (25 + 15) \times 6 = 120 \text{ cm}^2$

Thus, assertion (A) is true, but reason (R) does not give assertion (A).

Solution 37: (d) Assertion is false and Reason is true.

Clearly, reason (R) is true.

Assertion: Area of a parallelogram = base × height

$\Rightarrow AB \times DE = AD \times BF$

$\Rightarrow AD = (16 \times 8) \div 10 = 12.8 \text{ cm}$

So, the assertion is false.